Project Report

BY PIERRE QUEREUIL / ON MARCH 19, 2026 / IN PROJECT

Due: April 22nd, 2026 – NOTE: There will be no regrade window for this milestone.

Earning **1 or 2 late days on the project report** is based on the class evaluation (https://eval-duke.evaluationkit.com/) completion rate as follows:

- If 75% of the class completes evaluations, the class will earn 1 late day on the project report. So it will be due the day after LDOC.
- If 90% of the class completes evaluations, the class will earn an additional late day for the project report. So it will be due 2 days after LDOC.

Extra late days on the project report and video must be earned by **Monday, 4/20, 5pm**. You will get an announcement stating how many late days you all have earned at that time.

By default, you will not receive written feedback on the report and videos as you have on previous milestones. If your group would like to receive written feedback, please fill out this form: https://forms.office.com/r/9dUDQKVfSL.

Part 0: General Directions (5 points)

The final report is intended to be a comprehensive account of your collaborative data science course project. The report should demonstrate your ability to apply the data science skills you have learned to a real-world project holistically, from posing research questions and gathering data to analysis, visualization, interpretation, and communication. The report should stand on its own so that it makes sense to someone who has not read your proposal or prototype. This report may be one way your project group decides to publicly share your work if you agreed to do so during the initial plan. And if you disagreed then, but now want to, now is the time to discuss with your group!

The report should contain at least the parts defined below. In terms of length, it should be 5.5-7 pages using standard margins (1 in.), font (11-12 pt), and line spacing (1-1.5). A typical submission is around 3-4 pages of text and 5-7 pages overall with tables and figures. It is important to stay within the page limit, as practicing being succinct is an important skill. **Your final report should also have a descriptive title, not "CS216 Project Report".** You should convert your written report to a pdf and upload it to Gradescope under the assignment "Project Final Report" by the due date, and assign the appropriate pages to questions in the grading rubric. Be sure to include your names and NetIDs in your final document and use the group submission feature on Gradescope. You do not need to upload your accompanying data, code, or other supplemental resources demonstrating your work to Gradescope; instead, your report should contain instructions on how to access these resources (see the Results and Methods section below for more details).

In general, your approach to this report should be to write as if you had "planned this as your project all along." A report is not a chronological story of your project. It is a standalone summary of what you did, where the "story" serves the reader's comprehension. You should provide enough information that the results are understandable and your reader is convinced that the results are reasonable and accurate. An example project report is available in the class Box folder.

Grading

- **E (Exemplary, 5pts)** – Work that meets all requirements in terms of formatting and sections; report includes a descriptive title.
- **S (Satisfactory, 4 pts)** – Work that meets all requirements but is over 7 pages.

- **N (Not yet, 2 pts)** – Does not meet all requirements.
- **U (Unassessable, 1 pts)** – Missing at least one section.

Part 1: Introduction and Research Questions (15 points)

Your final report should begin by introducing your topic and restating your research question(s) as in your proposal. As before, your research question(s) should be (1) substantial, (2) feasible, and (3) relevant. In contrast to prior reports, **the final report does not need to explicitly justify in the text that the research questions are substantial and feasible**; your results should demonstrate both of these points. Therefore, *you should remove that text to save space*.

You should still explicitly justify how your research questions are relevant, although this should not be bolded as in previous milestones. In other words, be sure to explain the motivation of your research questions. Remember that relevant research questions address a subject of importance and interest within the scientific community or broader society. Additionally, we are looking for why your group believes this research project was worthwhile for your time **in this course**.

You can start with the text from your prototype, but you should update your introduction and research questions to reflect changes in or refinements of the project vision. You should not state specific updates. Rather, **write the report as the final product, and the prior milestones do not exist**. Pretend the readers are unaware of the prior milestones. If you feel an explanation of changes since the prototype is warranted, place that in the appendix. Your introduction should be sufficient to provide context for the rest of your report, and the content should be properly cited if it is drawn from external sources.

Grading

- **E (Exemplary, 15pts)** – Comprehensive introduction with clearly labeled, up-to-date research questions and a justification for how the research questions are relevant. Report introduction can stand alone without references to prior versions of the project; *no text for explicit justifications for "substantial" and "feasible" are made for the research question(s).*

- **S (Satisfactory, 14pts)** – Comprehensive introduction with clearly labeled, updated research questions and a justification for how the research questions are relevant. The introduction and research questions may not have been refined from the prototype (they have still kept reasoning for why their research questions are substantial and feasible).
- **N (Not yet, 9pts)** – Incomplete introduction where the research questions or justification are missing pieces, but at least some of it is present. Or the justification is clearly not reasonable.
- **U (Unassessable, 3pts)** – Incomplete introduction where it is entirely missing the research questions or justification or does not demonstrate meaningful effort.

Part 2: Data Sources (15 points)

Discuss the data you have collected and are using to answer your research questions. Be specific: name the datasets you are using, the information they contain, and where they were collected from / how they were prepared. You can begin with the text from your prototype, but be sure to update it to fit the vision for your final project. All data sources should be properly cited in this report by providing a working link or using a citation if you have a references section.

Grading

- **E (Exemplary, 15pts)** – Origins of all data are properly specified, cited, and relevant to answering the research question(s). If any significant data wrangling, cleaning, or other data preparation was done, these processes are explained. It is clear how the data was prepared for analysis.
- **S (Satisfactory, 14pts)** – Origins of all data are properly specified and cited. However, the justification is not clear why the data is relevant to the proposed research question(s). If any significant data wrangling, cleaning, or other data preparation was done, these processes are explained.
- **N (Not yet, 9pts)** – Poorly specified data sources or the justification for using that data set or the methods to acquire the data is lacking. No discussion of preparing the dataset.
- **U (Unassessable, 3pts)** – Data sources or methods to acquire data are missing or do not demonstrate meaningful effort.

Part 3: What Modules Are You Using? (15 points)

Your project should utilize concepts from modules we have covered in this course to answer your research question(s). We will assume you will use modules 1 (Python), 2 (Numpy/Pandas), and 5 (Probability). Your final report should state at least 3 more modules that you have utilized for your project. Each module should have a short description of how you used the knowledge in this module and a justification for that use. In addition, include what specific concepts from the module you used and at what stage of your project you mostly used this module. Potential stages include, but are not limited to: data gathering, data cleaning, data investigation, data analysis, and final report.

- Module 3: Visualization
- Module 4: Data Wrangling
- Module 6: Combining Data
- Module 7: Statistical Inference
- Module 8: Prediction & Supervised Machine Learning
- Module 9: Databases and SQL
- Module 10: Deep Learning

Your overall report should clearly show that you used the modules discussed in this section. You should add any additional modules used and update the existing modules to be more specific to the different tasks and stages of your projects that changed since your prototype.

Grading

- **E (Exemplary, 15pts)** – States at least 3 modules. For each module, they provide an updated (1) short description of how they used the module, (2) justification for using this module, (3) specific concepts they used, (4) what stage they used it, and (5) clearly implemented it in the final report.
- **S (Satisfactory, 14pts)** – States at least 3 modules. For each module, they provide an updated (1) short description of how they used the module, (2) justification for using this module, (3) what concepts they used and (4) what stage they used it. Less than 3 modules are clearly implemented in the final report.
- **N (Not yet, 9pts)** – States at least 3 modules. For each module, they provide an updated (1) short description of how they used the module, (2)

justification for using this module, (3) what concepts they used and (4) what stage they used it. Only one module is clearly implemented in the final report.

- **U (Unassessable, 3pts)** – Does not meet the Not Yet criteria.

Part 4: Results and Methods (15 points)

This is likely to be the longest section of your paper with multiple pages. The results and methods section of your report should explain your detailed results and the methods used to obtain them. Where possible, results should be summarized using **clearly labeled tables or figures** (i.e. Fig 1: XYZ) and supplemented with written explanations of the significance of the results with respect to the research questions outlined previously. **Please note that a screenshot of your dataset or a screenshot of a dataframe does not count as a table or figure and should not be included in your final report.**

The description of your methods should be specific. For example, if you scraped multiple web databases, merged them, and created a visualization, then you should explain how each step was conducted in enough detail that an informed reader could reasonably be expected to reproduce your results with time and effort. Just saying, "we cleaned the data and dealt with missing values" or "we built a predictive model" is insufficient.

Your report should also contain instructions on how to access your full implementation (that is, your code, data, and any other supplemental resources like additional charts or tables). The simplest way to do so is to include a link to the box folder, GitLab repo, or whatever other platforms your group is using to house your data and code. **Double-check that the link is accessible to anyone so that we can see it (e.g., check with incognito mode)!**

Grading

- **E (Exemplary, 15pts)** – Results are thoroughly discussed using clearly labeled tables or figures followed by written descriptions. Specific explanation of how the results were generated and from what data. Link to code/data to create charts or visualizations is provided.
- **S (Satisfactory, 14pts)** – Results are thoroughly discussed using clearly labeled tables or figures followed by written descriptions. Explanation of

how the results were generated may lack some specification or it is somewhat unclear as to what data the results are from. Link provided.

- **N (Not yet, 9pts)** – Results are discussed using tables with missing labels or lacking written descriptions. It is unclear how the results were generated and from what data.
- **U (Unassessable, 3pts)** – Results are missing or do not demonstrate meaningful effort.

Part 5: Limitations and Future Work (10 points)

In this part, you should discuss any important limitations or caveats to your results with respect to answering your research questions. For example, if you don't have as much data as you would like or are unable to fairly evaluate the performance of a predictive model, explain and contextualize those limitations. You may want to consider any **ethical implications or potential biases** of your results as well.

Finally, provide a brief discussion of future work. This could explain how future research might address the limitations you outline, or it could pose additional follow-up research questions based on your results so far. In short, explain how an informed reader (such as a peer in the class) could improve on and extend your results.

Grading

- **E (Exemplary, 10pts)** – Comprehensive and explicit discussion of important limitations and caveats to results. Brief discussion of future work and how results could be extended and improved upon.
- **S (Satisfactory, 9pts)** – Discussion of important limitations and caveats to results could be improved or the discussion of future work and how results could be extended and improved upon lacks some specification.
- **N (Not yet, 6pts)** – Incomplete discussion of important limitations and caveats to results. Discussion of future work and how results could be extended and improved upon may lack some specification.
- **U (Unassessable, 2pts)** – Limitations and future work are missing or do not demonstrate meaningful effort.

Part 6: Conclusion (10 points)

Provide a brief (one or two paragraphs) summary of your results. This summary of results should address all of your research questions.

Example

If one of your research questions was "Did COVID-19 result in bankruptcy in North Carolina during 2020?" then a possible (and purely hypothetical) summary of results might be:

We aggregate the public records disclosures of small businesses in North Carolina from January 2019 to December 2020 and find substantial evidence that COVID-19 did result in a moderate increase in bankruptcy during 2020. This increase is not geographically uniform and is concentrated during summer and fall 2020. We also examined the impact of federal stimulus but cannot provide an evaluation of its impact from the available data.

Grading

- **E (Exemplary, 10pts)** – Research questions are clearly and completely addressed through a summary of results.
- **S (Satisfactory, 9pts)** – Research questions are clearly addressed through a summary of results. The results may be lacking in completely answering the research questions.
- **N (Not yet, 6pts)** – Research questions are somewhat addressed through a summary of results. The results are lacking in completely answering the research questions. Or the results of one of the research questions is missing.
- **U (Unassessable, 2pts)** – Conclusion is missing or does not demonstrate meaningful effort.

Part 7: Collaboration Reflection & Sharing Plans (10 points)

This section should take about a half-page. In it you should pull out your collaboration plan and reflect on how the semester went, where it went well, and how it did not go well. This is an opportunity to think about how each of

you can be a more effective team member in future teams. In addition, confirm that you all still agree about how you want to share your project.

1. How much did your collaboration align with what was in the initial plan?
2. What went well in your collaboration? What did not go well?
3. What would you write differently in your collaboration plan if you did this again? Any questions you'd want outlined that are currently not in it?
4. What is your current plan for sharing your project publicly? Is it different from the original plan?

Grading

- **E (Exemplary, 10 pts)** – Collaboration reflection is communicated clearly and well thought out. Explicitly address post-semester sharing plans for the project.
- **S (Satisfactory, 9 pts)** – Collaboration reflection is missing depth or missing explicit mention of how the project will be shared (if at all).
- **N (Not yet, 6 pts)** – Reflection and project dissemination do not meet S.
- **U (Unassessable, 2 pts)** – Both are missing or do not demonstrate meaningful effort.

Part 8: AI Disclosure (5 points)

Just like homework, add a disclosure of how you used AI for this milestone. See Artificial Intelligence Disclosure (AID) Framework for how to write this disclosure. If you have difficulty explaining what you did, do your best and also include links to the chat logs. If no AI was used, simply write "Artificial Intelligence Tool(s): None"

Grading

- **E (Exemplary, 5pts)** – Work that meets all formatting requirements.
- **S (Satisfactory, 3pts)** – Does not meet all requirements.
- **0pts** – Blank

(Optional) Part 9: Appendix of additional figures, tables, and updates summary.

If you are struggling to keep your report within the page limit, you may move **some (not all)** of your figures and tables to an optional appendix that will not count against your page limit. **However, your report should stand on its own without the appendix**. The appendix is for adding more nuance to your results, not to give you more space to talk about your results. Succinctness is an important skill to practice when doing data science. **Your grader is not required to look at the appendix when grading.**

If you strongly feel like a summary of project updates since the proposal is required, you may put them in this appendix as well and mention they are in the appendix in the introduction.

Checklist Before You Submit:

1. Does your final report satisfy all general directions?
    1. 5.5-7 pages in length
    2. Standard margins (1 in.)
    3. Font size is 11-12 pt
    4. Line spacing is 1-1.5
    5. Final document is a pdf
    6. Descriptive project title
2. Do you have an Introduction and clearly stated Research Question(s)?
    1. Do you feel as if this part meets the requirements of **E (Exemplary)** or **S (Satisfactory)**?
3. Have you properly specified/cited one or more specific Data Sources and justified why they are relevant to the research Questions?
    1. Do you feel as if this part meets the requirements of **E (Exemplary)** or **S (Satisfactory)**?
4. Did you state at least 3 Modules to be used and how, as well as a justification of which concepts will be used at specific stages of the project?
    1. Do you feel as if this part meets the requirements of **E (Exemplary)** or **S (Satisfactory)**?

5. Have you reported all of your Results and Methods, including a specific explanation of how the results were generated?
    1. Do you feel as if this part meets the requirements of **E (Exemplary)** or **S (Satisfactory)**?
6. Does your report contain instructions on how to access your full implementation?
7. Have you defined clear Limitations to your results and Future Work?
    1. Do you feel as if this part meets the requirements of **E (Exemplary)** or **S (Satisfactory)**?
8. Have you written a comprehensive Conclusion?
    1. Do you feel as if this part meets the requirements of **E (Exemplary)** or **S (Satisfactory)**?
9. Have you written the Collaboration Reflection?
    1. Do you feel as if this part meets the requirements of **E (Exemplary)** or **S (Satisfactory)**?
10. Have you added your AI Disclosure?

PREVIOUS
Module 09: Databases and SQL

NEXT
Project Video Presentation

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